Part 21 — Against a reviewer's checklist: the five things that were thin

You asked me to adjust the module to a PI's stated tiers. I checked the module against every item on that list before writing anything. Most of it was already covered, in several cases heavily. Five things were not, and one of them sits in Tier 1.

The audit against the three tiers

Tier	Requirement	Coverage	Verdict
1	Central carbon, redox balancing, theoretical yields	307 mentions; parts 2, 14, 15, 18	Strong
1	CRISPR Cas9, multiplex, markerless, integration	47 and 58 mentions; part 7, PP5, PP6	Covered
1	**Cas12a, base editors, prime editing**	3 and 1 mentions	**Thin — chapter W1**
1	Enzyme kinetics, directed evolution, active-site engineering	Kinetics strong; evolution and hotspots 8 mentions	Adequate, strengthened in W1's closing
1	k_La, OUR, fed-batch, burden	174 mentions; parts 11, 19, 20	Strong
2	FBA, COBRApy, genome-scale models	42 mentions; part 10, MC1–MC5, Y1	Strong
2	Multi-omics literacy	154 mentions; part 4, chapter S5, Z5, Z6, Z9	Strong
2	AlphaFold, ESM, ML-guided evolution	14 and 38 mentions; part 8, P1, P2, Y5	Adequate
2	Biosensors	33 mentions; part 8 ch 26	Covered
2	**FACS and droplet microfluidics**	4 and 10 mentions	**Thin — chapter W3**
3	Python automation, NGS pipelines, statistical DoE	Heavy; part 20 chapters V4, DE1–DE4	Strong
3	Non-conventional chassis: autotrophs, methanotrophs, thermophiles	42 mentions; part 6, X4	Covered
3	Techno-economic assessment	Part 4 ch 49, G1, X22, PP12	Strong
3	**Life cycle analysis**	7 mentions	**Thin — chapter W4**
—	Named products: terpenoids, biopolymers, rare amino acids	39, 568, 45 mentions	Strong
—	**Named products: flavonoids and polyphenols**	5 mentions	**Thin — chapter W2**

Four chapters, W1 to W4, and eight problems, EV1 to EV8, taking the set to **141**. One observation worth making: the reviewer's list is close to what the module already contained, which is a reasonable sign that the curriculum was well aimed. The gaps are specific rather than structural.

Chapter W1 · extends weeks 23 and 24 · Tier 1

The editing toolkit in full: Cas9, Cas12a, base editors, prime editing

Part 7 taught Cas9 and PP5 taught the counterselection arithmetic. A reviewer expects you to know which editor for which job, and that is a question about PAM, editing window and what repair pathway the host actually has.

The four tools, compared on what decides the choice

	Cas9	Cas12a (Cpf1)	Base editors	Prime editing
PAM	5′-NGG-3′, 3′ of the protospacer	5′-TTTV-3′, 5′ of it	Inherited from the Cas it is built on	Same
Cut	Blunt, 3 bp from the PAM	**Staggered, 4–5 nt overhangs**, distal to the PAM	Nick or none	Nick on one strand
Guide	crRNA + tracrRNA, or a 100-nt sgRNA	**crRNA alone, ~43 nt**	As the parent Cas	**pegRNA**: guide + primer binding site + reverse-transcription template
Multiplexing	One guide per promoter, or an array with processing	**Self-processes its own array** — one transcript, many guides	As parent	Hard: pegRNAs are long
Makes	A double-strand break	A double-strand break	**C→T (CBE) or A→G (ABE)**, no break	**Any substitution, small insertion or deletion**
Needs a donor?	Yes, in bacteria	Yes	**No**	No — the template is in the pegRNA

The fact that governs everything in bacteria: most bacteria have no efficient non-homologous end joining, so a double-strand break is **lethal, not mutagenic**. That is why Cas9 and Cas12a in *E. coli* are used as a *counterselection* — recombineering supplies the edit and the nuclease kills the unedited, which is PP5's arithmetic. In organisms that do have end joining (many actinomycetes and mycobacteria) the same cut gives small indels directly, and the strategy is entirely different. Check whether your host has *ku* and *ligD* before choosing.

Where each one wins

Cas12a for multiplexing. Because it processes its own CRISPR array, a single transcript can carry six or ten guides, where Cas9 needs a promoter or a processing system per guide. Its AT-rich PAM also complements Cas9's GC-rich one, which matters in a high-AT or high-GC genome where NGG sites may be scarce near your target. Its staggered cut leaves overhangs that favour insertion of a matching donor.

Base editors for point mutations at scale and for essential genes. A cytosine base editor makes C→T within a window of roughly positions 4 to 8 of the protospacer; an adenine base editor makes A→G similarly. No break, no donor, no counterselection — which is why base editing is the tool for making a **premature stop codon** (CAA, CAG, CGA → TAA, TAG, TGA) to knock a gene out without deleting it, and for installing a point mutation in an essential gene where a deletion would be lethal. The constraints are real and arithmetic: the target base must fall in the window, which means a PAM at the right distance; and **bystander editing** of other C's or A's in the window is the failure mode — a window containing three cytosines gives a mixture of edited products, which EV2 computes.

Prime editing for anything the other three cannot do: a substitution outside a base editor's chemistry, a small insertion, a short deletion, or a precise change with no bystanders. The cost is efficiency and complexity — the pegRNA must be designed with a primer binding site and template whose lengths are tuned empirically, and reported efficiencies in bacteria are low enough that a selection or screen is usually still needed.

Choosing, as a decision list

What you need	Tool	Why
Clean deletion of a gene or operon	λ-Red or RecET + Cas9 counterselection	Recombineering makes it, the nuclease enriches it (PP5, PP6)
Six knockouts at once	Cas12a array, or MAGE	Self-processing array; or MAGE's per-cycle arithmetic (PP5)
Knock out an essential gene's function	CBE, making a stop codon	No break, no lethality; or CRISPRi for knockdown
One point mutation, no scar	Base editor if in-window, else prime editing	Window and PAM decide it (EV2)

Tune expression without editing	CRISPRi with dCas9	Titratable, reversible, multiplexable
Insert a 3 kb cassette	Recombineering with long homology, or integrase	Beyond prime editing's size range

Two honest limits. Reported efficiencies for base and prime editing in non-model bacteria are **recalled and variable** — verify against a recent review for your host before planning a campaign. And the choice between editors is often settled by what is already working in the laboratory you join, which is a legitimate reason and worth saying out loud in an interview.

Chapter W2 · extends weeks 53 to 55 · the named products

Flavonoids: a second aromatic pathway with a different bottleneck

The module did terpenoids, polymers and amino acids. Flavonoids are the fourth product class a reviewer named, and they fail for a reason the other three do not: malonyl-CoA.

The pathway, and the yield ceiling

Naringenin ($C_{15}H_{12}O_5$, 272.26) is the branch point to most flavonoids. The route is: tyrosine → *p*-coumarate (tyrosine ammonia lyase, TAL) → *p*-coumaroyl-CoA (4-coumarate-CoA ligase) → condensation with **three malonyl-CoA** by chalcone synthase → naringenin chalcone → naringenin (chalcone isomerase). Downstream, flavonols, anthocyanins and so on come from oxidations and glycosylations, most of them by cytochrome P450s.

The carbon accounting has two tributaries. The C_9 aromatic ring comes from tyrosine, which needs erythrose-4-phosphate (4 C) plus two PEP (6 C), losing one CO_2 in the shikimate route — so **1.67 glucose per ring**. The remaining six carbons come from three acetyl units via malonyl-CoA, which is three pyruvate losing three CO_2 at pyruvate dehydrogenase — **1.5 glucose**. Total 3.17 glucose per naringenin, giving a mass yield ceiling of

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272.26 / (3.17 × 180.156) = 0.477 g/g
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Reported titers are typically tens to hundreds of milligrams per litre†, so a 0.5 g/L result from 20 g/L of glucose is about **5% of the ceiling** — an order of magnitude below where terpenoids and amino acids sit, and that gap is the subject.

Why malonyl-CoA is the bottleneck

Malonyl-CoA is made by acetyl-CoA carboxylase and consumed almost entirely by fatty-acid synthesis, where it is kept at a very low pool and tightly regulated. A flavonoid pathway needs three of them per product, and it is competing with the cell's own membrane synthesis for a metabolite that the cell deliberately does not accumulate. The demand arithmetic: 5 g/L of naringenin over 48 hours needs **55 mmol/L of malonyl-CoA**, which at 10 gDW/L is **0.115 mmol·gDW^{-1}·h^{-1}** — comparable with the cell's own fatty-acid demand of roughly 0.2. You are asking the cell to roughly double its malonyl-CoA flux and divert the increase.

The interventions, in the playbook form of part 20 chapter V3: **overexpress acetyl-CoA carboxylase** (the direct supply route, with a biotin requirement and an ATP cost); **inhibit fatty-acid synthesis** with cerulenin, which works in a flask and is far too expensive at scale, or by titrating *fabB* or *fabF* with CRISPRi, which is the scalable version; **raise acetyl-CoA** by the routes of part 6 chapter 58, including phosphoketolase; and **stage the process**, growing first and diverting malonyl-CoA later, which is PP8's two-stage calculation applied to a precursor rather than a product.

The second problem: cytochrome P450s

Most flavonoid decorations are P450 oxidations, and P450s are hard in bacteria for three compounding reasons. They are **membrane-anchored** in plants, so expression in *E. coli* needs truncation or a fusion and typically loses an order of magnitude or more of activity†. They need a **redox partner** — a cytochrome P450 reductase delivering two electrons from NADPH per turnover — which must be co-expressed at a matched ratio. And they **uncouple**: when the electron flow is not coupled to substrate oxidation, the electrons leak to peroxide or water, so the NADPH cost per product rises. At 100% coupling a turnover costs 2 NADPH; at 50% it costs 4; at **20% coupling it costs 10**, and a three-step P450 cascade at that coupling costs **30 NADPH per molecule of product** — which is why a decorated flavonoid is far harder than naringenin itself, and why yeast, with native ER membranes and a native reductase, is often the better host for the decorated compounds even though *E. coli* makes the core faster.

Naringenin ceiling	3.17 glucose per molecule → 0.477 g/g; reported titers ≈5% of it
Malonyl-CoA demand	3 per product; 5 g/L needs 0.115 mmol·gDW^{-1}·h^{-1} against ~0.2 for fatty acids
P450 cost	2 NADPH per turnover at full coupling; 10 at 20%; 30 for a three-step cascade
Host logic	*E. coli* for the core skeleton, yeast for the P450 decorations

What to open a book for. A review of microbial flavonoid production for the titers and the malonyl-CoA strategies — and read it specifically for what each group did about malonyl-CoA, which is the only question that separates the papers. For P450s, a review of their functional expression in microbial hosts.

Chapter W3 · extends weeks 29 and 30 · Tier 2

Sorting and droplets: screening at 10^7 instead of 10^3

A plate reader screens thousands. A sorter screens hundreds of millions. The arithmetic of what that buys — and of what it costs in false positives — is Tier 2 material and the module had four mentions of it.

FACS: throughput and enrichment

A cell sorter processes events at kilohertz to tens of kilohertz. At 5,000 events per second a four-hour run sees **72 million cells**; at 20,000 it sees **288 million**; at 50,000, **720 million**. Against problem X23's campaign of 387 strains and part 8 chapter 29's library of 30,000, that is a different regime entirely — the constraint stops being throughput and becomes **whether the thing you want is visible as fluorescence**.

Enrichment per round is the Bayes calculation. With a true-positive frequency f and a sort that is correct a fraction p of the time, the post-sort frequency is fp/(fp + (1−f)(1−p)):

Starting frequency	Purity 0.90	Purity 0.99
10^{-5}	9.0×10^{-5} (9×)	**9.9×10^{-4} (99×)**
10^{-4}	9.0×10^{-4} (9×)	9.8×10^{-3} (98×)
10^{-3}	8.9×10^{-3} (9×)	9.0×10^{-2} (90×)

Two readings. Enrichment per round is roughly p/(1−p) and **almost independent of the starting frequency** — which is why rounds compound: two rounds at 99% purity take 10^{-5} to **8.9%**, and three take it to dominance. And **purity matters far more than throughput**: going from 90% to 99% is a tenfold gain per round, where doubling the event rate is a twofold gain once.

The hard part is not the machine. It is **linking the product to fluorescence**: a transcription-factor biosensor (part 8 chapter 26), an enzyme-coupled reaction producing a fluorophore, or a FRET sensor. Each brings its own failure modes — the sensor can be gamed by a mutation that activates it directly rather than by making product, which is the single most common way a sorted library yields nothing. A **negative control strain carrying the sensor and no pathway**, sorted alongside, is what detects that.

Droplets: Poisson loading and what it buys

Water-in-oil droplets give each cell a private compartment, so a **secreted** product stays with its producer — which FACS cannot do, since a secreted product diffuses away. Loading follows Poisson statistics at mean occupancy λ:

λ	Empty	Exactly one cell	More than one	Singles as a share of occupied
0.1	90.5%	9.0%	0.5%	**95.1%**
0.3	74.1%	22.2%	3.7%	85.7%
1.0	36.8%	36.8%	26.4%	58.2%

The trade is between wasting droplets and corrupting them. At λ = 0.1 you discard 90% of your droplets as empty but **95% of occupied droplets hold exactly one cell**, so a positive droplet identifies a single genotype. At λ = 1.0 you use the droplets efficiently and 26% hold more than one cell, so a positive result is ambiguous and a strong producer can carry a passenger. For a *screening* experiment where you will re-isolate and re-test, λ near 0.3 is a reasonable compromise; for a *selection* where the droplet content is pooled and sequenced, λ must be low.

Volumes make it practical: 10^7 droplets of 50 pL is only **0.5 mL of emulsion**, which is why droplet screening at 10^7 is a bench-scale experiment while the equivalent in microtitre plates is 100,000 plates. What it costs is a bespoke device, a fluorogenic assay that works inside a droplet, and considerable trouble-shooting — which is why it belongs in Tier 2 rather than Tier 1.

FACS throughput	`72–720 million cells in a four-hour run at 5–50 kHz`
Enrichment	`fp/(fp + (1−f)(1−p)) ≈ p/(1−p) per round; 99× at 99% purity`
Droplet loading	`Poisson: λ = 0.1 gives 95% single-occupancy among occupied drops`
Scale	`$10^7$ droplets at 50 pL = 0.5 mL of emulsion`

What to open a book for. A review of biosensor-driven high-throughput screening in metabolic engineering; and, for droplets, a methods review on microfluidic droplet screening for enzymes — read both for the failure modes rather than the throughput claims.

Chapter W4 · extends week 49 · Tier 3

Life cycle analysis beside the techno-economic model

The module costed processes in dollars. A reviewer in industrial biotechnology expects you to cost them in carbon too — and the two calculations use the same inputs and reach different conclusions.

The four things that define an LCA

The functional unit — what you are comparing. "1 kg of product at the factory gate" is the usual cradle-to-gate choice; cradle-to-grave adds use and disposal, which for a biopolymer is where the interesting part is. **The system boundary** — whether you include agriculture, land-use change, transport, utilities, waste treatment. **The impact categories** — global warming potential is the one everyone reports, but eutrophication and water use frequently decide against a bio-route, because agriculture drives both. And **allocation** — how burden is split between a product and its co-products, by mass, energy content or economic value, which is the single most contestable choice in any LCA.

The calculation, with the module's own numbers

Taking illustrative factors of 0.6 kg CO_2e per kg of glucose, 0.4 per kWh of electricity and 0.07 per MJ of steam†, and PP12's propanediol process:

Case	Sugar	Steam for water removal	Power	Total
Y 0.40, titer 50 g/L	2.50 kg (1.50)	42.9 MJ (3.01)	2.0 kWh (0.80)	**5.31 kg CO_2e/kg**
Y 0.40, titer 135 g/L	2.50 kg (1.50)	14.5 MJ (1.01)	2.0 kWh (0.80)	**3.31**
Y 0.51, titer 135 g/L	1.96 kg (1.18)	14.5 MJ (1.01)	2.0 kWh (0.80)	**2.99**

Read the first two rows together. **Raising titer from 50 to 135 g/L cuts the carbon footprint by 38%** while changing nothing about the strain's yield — because at low titer the steam to evaporate water is the largest single term, larger than the sugar. That is the same conclusion PP12 reached in dollars, reached independently in carbon, and it is the most useful thing an engineer can take from an LCA: **titer is a sustainability variable, not just an economic one**.

Against a fossil benchmark of roughly 2 to 4 kg CO_2e per kg for a commodity chemical†, the low-titer case *loses* and the high-titer case is competitive. A bio-process is not automatically lower-carbon, and saying so in an interview is more convincing than assuming the opposite.

The two arguments that change the answer

Biogenic carbon. The product contains carbon that was fixed from the atmosphere. For a $C_3H_8O_2$ product that is **1.74 kg of CO_2 per kg embodied** in the molecule, and whether you may claim it depends on the end of life: if the product is burned or composted it returns, and a cradle-to-grave analysis cancels it; if it is a durable polymer in long-term use it is genuine sequestration for that period. This is where biopolymer LCAs are won and lost, and where they are most often overstated.

Allocation. A process yielding 1 kg of product and 0.5 kg of animal feed can assign the upstream burden either entirely to the product or partly to the feed. Allocating a third to the co-product takes the 135 g/L case from 3.31 to **2.82 kg CO_2e/kg**, and allocating half takes it to **2.56** — a 23% improvement from a bookkeeping decision with no change to the process at all. Hence the ISO guidance to *avoid* allocation where possible by expanding the system boundary, and the practical requirement to report results under more than one allocation rule so a reader can see the sensitivity.

Definition	`functional unit, system boundary, impact categories, allocation`
Titer effect	`50 → 135 g/L cuts GWP 38%, through steam rather than sugar`
Benchmark	`fossil commodity 2–4 kg CO₂e/kg: bio wins only at high titer`
Biogenic carbon	`1.74 kg CO₂ per kg of a C₃H₈O₂ product; claimable only with a durable end of life`
Allocation	`33% to a co-product moves 3.31 → 2.82 kg CO₂e/kg with no process change`

What to open a book for. ISO 14040 and 14044 for the framework, skimmed for the definitions rather than read; a published LCA of a bio-based chemical for how a real boundary is drawn; and the ecoinvent or GREET documentation for where emission factors come from — the factors above are illustrative and should be replaced by sourced ones.

Problems EV1 to EV8

EV1 · chapter W1 · 1.5 h

Choose the editor for six jobs

1. For each of six tasks — a clean operon deletion, six simultaneous knockouts, inactivating an essential gene, one scarless point mutation, tunable knockdown, and a 3 kb insertion — name the tool and justify it.
2. Which two tasks change answer completely if the host has *ku* and *ligD*?
3. Why is a double-strand break lethal rather than mutagenic in *E. coli*?
4. Your target has no NGG PAM within 10 bp. What are your three options?

Worked solution. Operon deletion: recombineering with Cas9 counterselection. Six knockouts: a Cas12a array (self-processing) or MAGE. Essential gene: a cytosine base editor making a stop codon, or CRISPRi for knockdown rather than knockout. Scarless point mutation: base editor if the target falls in the window, otherwise prime editing. Tunable knockdown: CRISPRi with dCas9. 3 kb insertion: recombineering with long homology arms or a site-specific integrase — beyond prime editing's size range.

The first two change completely with *ku* and *ligD*: in an end-joining host a cut alone produces indels, so you can knock out a gene with a guide and no donor at all. A break is lethal in *E. coli* because there is no end-joining pathway and homologous recombination needs a template — without one the chromosome is broken irreparably. With no NGG nearby: use **Cas12a** and its TTTV PAM, use a **PAM-relaxed Cas9 variant**, or **move the edit** — a synonymous change a few bases away is often as good, and is how domestication is done anyway (part 7 chapter 21).

What it teaches: editor choice is decided by PAM availability, repair pathways present in the host, and whether a break can be tolerated — not by which enzyme is newest.

EV2 · chapter W1 · 1.5 h

Base editing: windows, stop codons and bystanders

1. A cytosine base editor edits C→T at protospacer positions 4–8. Your target C is at position 6, and there are C's at positions 4 and 9. What products do you expect?
2. Which codons can a CBE turn into a stop, and which into a non-stop missense?
3. If each in-window C is edited independently with probability 0.6, what fraction of molecules carry the intended edit alone?
4. How would you remove the bystander problem without changing the editor?

Worked solution. The C at position 9 is outside the window and should be untouched; the C at position 4 is inside it and is a bystander. With independent editing at p = 0.6, the fraction carrying the target edit *and not* the bystander is 0.6 × 0.4 = **24%**, against 36% carrying both, 24% carrying only the bystander and 16% unedited. So **the single most likely outcome is the double edit**, and screening must sequence rather than assume.

A CBE makes stops from **CAA → TAA, CAG → TAG and CGA → TGA** — all on the sense strand, plus the reverse-complement equivalents when the guide targets the other strand, which doubles your options. Without changing the editor, the bystander problem is solved by **choosing a different guide** whose window places the target C alone, which is almost always possible given the PAM density, or by exploiting the genetic code: a bystander that is **synonymous** does not matter, so check the reading frame before rejecting the guide.

What it teaches: base editing is a window problem, and the guide is chosen by what else falls in the window, not just by what you want to change.

EV3 · chapter W2 · 2 h

The naringenin ceiling and where the gap is

1. Account for the carbon in naringenin from both tributaries and compute the glucose requirement.
2. Give the mass yield ceiling and express a 0.5 g/L titer from 20 g/L glucose as a fraction of it.
3. Compute the malonyl-CoA demand for 1 and 5 g/L over 48 h at 10 gDW/L, against a fatty-acid demand of 0.2 mmol·gDW^{-1}·h^{-1}.
4. Rank four interventions and say which you would test first.

Worked solution. The C_9 ring costs 1.67 glucose (erythrose-4-phosphate plus two PEP, one CO_2 lost in shikimate) and

the three acetyl units cost 1.5 (three pyruvate, three CO_2 lost at pyruvate dehydrogenase): **3.17 glucose per naringenin**, a ceiling of **0.477 g/g**. A 0.5 g/L titer from 20 g/L is 0.025 g/g, which is **5.2% of the ceiling** — the widest gap of any product in this module.

Malonyl-CoA demand is **0.023 mmol·gDW^{-1}·h^{-1} at 1 g/L and 0.115 at 5 g/L**, against roughly 0.2 for the cell's own fatty acids: at 5 g/L you are asking for a 57% increase in total malonyl-CoA flux, all of it diverted. Ranked: **titrate fabB/fabF with CRISPRi first** — it is reversible, tunable and tests the hypothesis that malonyl-CoA is limiting in one experiment; then **overexpress acetyl-CoA carboxylase**, which adds supply and costs ATP and biotin; then **raise acetyl-CoA** by phosphoketolase or the routes of part 6 chapter 58; and **stage the process** last, since that is architectural. Testing CRISPRi first is the cheap diagnostic: if knocking down fatty-acid synthesis does not move the titer, malonyl-CoA was not the limit and the other three are wasted work.

What it teaches: a product can be carbon-cheap and precursor-poor, and the first experiment should test which of the two the limitation actually is.

EV4 · chapter W2 · 1.5 h

What a P450 cascade costs

1. Compute NADPH per turnover at coupling efficiencies of 100%, 50% and 20%.
2. A decorated flavonoid needs three P450 steps at 20% coupling. What is the NADPH bill per product?
3. Compare that with the biomass NADPH demand of 13.03 mmol/gDW from part 14 chapter B5.
4. Give three reasons P450s underperform in *E. coli*, and say when you would switch host.

Worked solution. Two electrons per turnover means **2 NADPH at full coupling, 4 at 50% and 10 at 20%**. A three-step cascade at 20% coupling is **30 NADPH per product molecule**. For perspective, making a gram of biomass costs 13.03 mmol of NADPH; at 1 mmol/L of product (0.27 g/L of a 272 Da compound) the cascade would consume 30 mmol/L — more than twice the NADPH in a gram of cells per litre, which makes the cofactor supply the binding constraint rather than carbon.

Three reasons for underperformance: **membrane anchoring**, since a plant P450 is ER-bound and must be truncated or fused for *E. coli*, usually losing activity; **redox partner mismatch**, since the native reductase is absent and a co-expressed one must be ratio-matched; and **uncoupling**, which both wastes NADPH and generates peroxide that damages the enzyme. Switch to yeast when **two or more P450 steps are required** — it has ER membranes, a native reductase and the trafficking machinery, and the loss in growth rate and genetic tractability is worth it at that point.

What it teaches: a P450's cost is set by its coupling efficiency, and an uncoupled cascade turns a carbon problem into a cofactor problem.

EV5 · chapter W3 · 1.5 h

Sorting: how many rounds, and what can fool you

1. How many cells does a 4-hour sort see at 5,000, 20,000 and 50,000 events per second?
2. Compute the post-sort frequency and enrichment from 10^{-5}, 10^{-4} and 10^{-3} at purities of 0.90 and 0.99.
3. Where does two rounds at 99% purity take a 10^{-5} variant?
4. Name the commonest way a sorted library yields nothing, and the control that detects it.

Worked solution. 72, 288 and 720 million cells in four hours. Enrichment is about 9× at 90% purity and about **99× at 99%**, almost independent of the starting frequency — so purity, not throughput, is the lever: a tenfold gain per round from improving the gate, against a twofold gain from doubling the event rate. Two rounds at 99% take 10^{-5} to **8.9%**, and a third makes it the majority.

The commonest failure is **sensor gaming**: a mutation that activates the reporter directly — in the promoter, the transcription factor, or the fluorescent protein — is selected far more easily than one that raises product, because it needs no metabolic change. The control that detects it is a **sensor-only strain** with no pathway, sorted in the same run: anything it yields is a false positive by construction. The follow-up is to re-test sorted clones for *product* by chromatography, not for fluorescence, which is the X17 and R1 measurement.

What it teaches: sorting enrichment compounds with purity, and the biology that defeats it is a sensor that can be satisfied without making product.

EV6 · chapter W3 · 1 h

Droplets: choose the loading

1. At mean occupancies of 0.1, 0.3 and 1.0, compute the empty, single and multiple fractions and the single share of occupied droplets.
2. Which λ for a screen you will re-isolate from, and which for a pooled selection you will sequence?
3. Compute the emulsion volume for 10^7 droplets at 50 and 20 pL.
4. What can a droplet screen do that FACS cannot, and at what cost?

Worked solution. At λ = 0.1: 90.5% empty, 9.0% single, 0.5% multiple, and **95.1% of occupied droplets hold exactly one cell**. At 0.3: 74.1 / 22.2 / 3.7, with 85.7% singles. At 1.0: 36.8 / 36.8 / 26.4, with only 58.2% singles.

For a screen you will re-isolate and re-test, **λ ≈ 0.3** is a fair compromise — you use the droplets and can tolerate some ambiguity because the hit is verified afterwards. For a **pooled selection you will sequence**, λ must be low (0.1 or below), because a multiply-occupied droplet assigns a strong phenotype to a passenger genotype and that error is unrecoverable. Volumes: 10^7 droplets is **0.5 mL at 50 pL and 0.2 mL at 20 pL** — bench-scale, against 100,000 microtitre plates for the same count.

What droplets add over FACS is compartmentalisation: a **secreted** product stays with its producer, so you can screen for secreted enzymes and exported metabolites that FACS cannot see. The cost is a bespoke device, a fluorogenic assay that works in a 50 pL drop, and the trouble-shooting that goes with both.

What it teaches: droplet loading is a deliberate trade between wasted droplets and ambiguous genotypes, and the right λ depends on whether you will verify the hit afterwards.

EV7 · chapter W4 · 1.5 h

The carbon footprint of your own process

1. Using 0.6 kg CO_2e per kg glucose, 0.4 per kWh and 0.07 per MJ of steam, compute the footprint per kg of product at (0.40 g/g, 50 g/L), (0.40, 135) and (0.51, 135).
2. Which term dominates at low titer, and which at high?
3. Compare with a fossil benchmark of 2–4 kg CO_2e/kg.
4. What does this say about where to spend engineering effort?

Worked solution. 5.31, 3.31 and 2.99 kg CO_2e per kg. At 50 g/L the steam to evaporate water is 3.01 of the 5.31 — **the largest single term, bigger than the sugar**; at 135 g/L it falls to 1.01 and sugar becomes dominant. So raising titer from 50 to 135 g/L cuts the footprint **38%** with no change in yield.

Against 2–4 kg CO_2e/kg for the fossil route, the low-titer case **loses outright** and the high-titer case is competitive. A bio-process is not automatically lower-carbon, and the honest version of the claim is conditional on titer. On effort: titer improvements pay twice, in separation cost and in carbon, which is the same conclusion PP12 reached in dollars; yield improvements pay once, in sugar; and decarbonising the *utilities* — the 0.4 kg CO_2e per kWh — is often larger than anything the strain can deliver, and is outside the laboratory's control, which is worth knowing before claiming otherwise.

What it teaches: the carbon calculation uses the same inputs as the economic one and weights them differently, with water removal dominating at low titer.

EV8 · chapter W4 · 1 h

Allocation and biogenic carbon: the two arguments that move the number

1. A process gives 1 kg of product and 0.5 kg of animal feed. Compute the product footprint when 0%, 33% and 50% of upstream burden is allocated to the co-product.
2. Compute the biogenic carbon embodied in 1 kg of a $C_3H_8O_2$ product.
3. When may that carbon be claimed as a credit, and when not?
4. What does ISO guidance say to do about allocation, and what would you report?

Worked solution. From the 3.31 kg CO_2e baseline: **3.31, 2.82 and 2.56 kg CO_2e/kg** at the three allocations — a **23% improvement from a bookkeeping decision** with no change to the process. That is why allocation is the most contested choice in any LCA and why two studies of the same process can differ by more than any engineering intervention would deliver.

Biogenic carbon: a $C_3H_8O_2$ product of molar mass 76.09 containing three carbons embodies 3 × 44.01/76.09 = **1.74 kg of CO_2 per kg**. It may be claimed as sequestration **only if the end of life keeps it out of the atmosphere** — a durable polymer in long service, or something landfilled without degradation. If the product is burned, composted or

consumed, a cradle-to-grave analysis returns it and the credit vanishes, which is the commonest overstatement in biopolymer marketing.

ISO 14044 says to **avoid allocation where possible**, by subdividing the process or expanding the system boundary to include the function the co-product displaces, and only then to allocate — by physical relationship first, economic value last. What I would report: the result under **system expansion as the base case and under mass and economic allocation as sensitivities**, with the three numbers side by side, so a reader sees that the conclusion survives the choice or does not.

What it teaches: two defensible bookkeeping choices move an LCA result by more than most process improvements, so a result reported under one rule is not a result.

Closing notes to part 21

What the reviewer's list changed, and what it did not

Most of the list was already covered, several items heavily. That is worth stating plainly rather than pretending to a bigger overhaul: the curriculum you brought was well aimed at this kind of laboratory. What the audit found was five specific thin spots, now filled:

Gap	Tier	Now in
Cas12a, base editors, prime editing	**1**	Chapter W1, problems EV1, EV2
FACS and droplet microfluidics	2	Chapter W3, problems EV5, EV6
Life cycle analysis	3	Chapter W4, problems EV7, EV8
Flavonoids and P450s	product class	Chapter W2, problems EV3, EV4
Reading list organised by tier	all	`me-reading-list-by-tier.pdf`

One disagreement worth recording

The reviewer puts techno-economic assessment and non-conventional chassis in Tier 3, "easily learned in the first year". The module's own arithmetic suggests TEA belongs higher: problem G1 showed that the titer–separation relationship determines whether a project is viable at all, and X22 showed that the input dominating the uncertainty is the one a laboratory does not control. A candidate who can say "this strain cannot reach a viable selling price at any plausible yield, because the water removal costs more than the product sells for" is making an argument no amount of first-year catch-up substitutes for. I would treat Tier 3's TEA as Tier 2, and the module already does.

On the numbers in this part

All computed here. Recalled and flagged: the reported flavonoid titers, the magnitude of P450 activity loss in bacteria, the emission factors for glucose, electricity and steam, the fossil benchmark, and the reported efficiencies of base and prime editing in bacteria. The emission factors in particular are illustrative placeholders — a real LCA sources them from ecoinvent or GREET, and the conclusion about titer survives any plausible change to them, which is the useful kind of robustness.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–20	77
Part 21 EV1–EV8	8
Total in the tracker	**141**